# Total Least Squares as a quadratic minimization problem

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**Abstract**

Given a set of points in $\mathbb{R}^d$, the Total Least Squares (TLS) problem aims at finding the affine hyperplane $L \subset \mathbb{R}^d$ such that the sum of the squared orthogonal distances of the points w.r.t. to $L$ is minimized. In this note, we show that the energy to be minimized is a quadratic function of the unitary vector orthogonal to the hyperplane and the matrix defining this quadratic function is the covariance matrix of the points. Therefore, the unitary vector orthogonal to the TLS hyperplane turns out to be the eigenvector of the covariance matrix which is associated to the lowest eigenvalue.

## Problem formulation

Let us consider a unitary vector $r \in S^d := \{x \in \mathbb{R}^d : |r| = 1\}$ and a real number $k \in \mathbb{R}$. We denote by $L_{r,k} \subset \mathbb{R}^d$ the set of points $x \in \mathbb{R}^d$ whose orthogonal projection over $r$ equals $k \in \mathbb{R}$. This set is called the affine hyperplane of $\mathbb{R}^d$ which is orthogonal to the versor $r$ with displacement $k$. Formally, it is defined as $L_{r,k} := \{x \in \mathbb{R}^d : x^T r = k\}$. Given any $a \in L_{r,k}$, by definition we have that $a^T r = k$. Therefore, we can also define

$$L_{r,a} := L_{r,k} = \{x \in \mathbb{R}^d : x^T r - a^T r = 0\} \tag{1}$$

and this definition clearly does not depend on the choice of $a \in \mathbb{R}^d$ such that $a^T r = k$. Given any point $z \in \mathbb{R}^d$, the orthogonal distance between $z$ and the affine hyperplane $L_{r,a}$ is the absolute value of the difference between the orthogonal projection of $z$ over $r$ and $k$, i.e.,

$$\operatorname{dist}(z, L_{r,a}) = |z^T r - k| = |z^T r - a^T r| = |(z-a)^T r| \tag{2}$$

for any $a \in L_{r,k}$. This situation is illustrated in Figure 1 in the particular case of $d = 3$. We can now state our problem.

**Problem** (Total Least Squares)**.** *Given a set of points $Z = \{z_j\}_{j=1}^N \subset \mathbb{R}^d$, find the affine hyperplane $L_{r,a}$ that minimizes*

$$E_Z(r, a) := \sum_{j=1}^{N} \operatorname{dist}(z_j, L_{r,a})^2. \tag{3}$$

*Equivalently, find $r^* \in S^d$ and $a^* \in \mathbb{R}^d$ such that*

$$(r^*, a^*) = \operatorname*{argmin}_{(r,a) \in S^d \times \mathbb{R}^d} E_Z(r, a). \tag{4}$$

*We will call $L_{r^*,a^*}$ the TLS hyperplane approximating the points $Z$.*

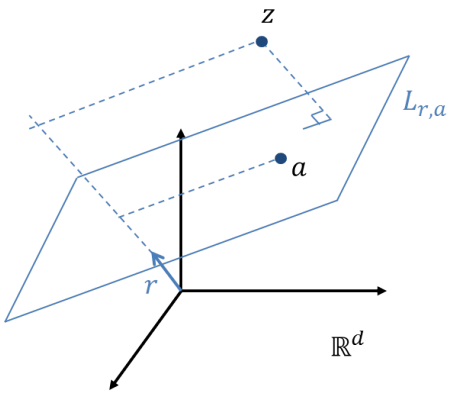



Figure 1: The orthogonal distance of the point $z$ to the affine plane $L_{r,a}$ can be obtained via scalar product as $|(z-a)^T r|$

## Equivalent matricial formulation of the TLS problem

In view of (2), the objective function to be minimized, i.e., the sum of the squared orthogonal distances of points $Z = \{z_j\}_{j=1}^N \subset \mathbb{R}^d$ w.r.t. the affine hyperplane $L_{r,a} \subset \mathbb{R}^d$, can be rewritten as

$$E_Z(r,a) = \sum_{j=1}^{N} \mathrm{dist}(z_j, L_{r,a})^2 = \sum_{j=1}^{N} |(z_j - a)^T r|^2. \tag{5}$$

By arranging the $N$ points $z_j \in \mathbb{R}^d$ in a $N \times d$ matrix (the $k$-th row of the matrix $Z$ is $z_j^T$) and by denoting $\mathbb{1} = (1, \ldots, 1)^T$ the $N \times 1$ vector with all entries 1, we can write the objective energy $E_Z(r,a)$ in a convenient matricial form as

$$\begin{aligned} E_Z(r,a) &= \sum_{j=1}^{N} |(z_j - a)^T r|^2 \\ &= r^T (Z - \mathbb{1}a^T)^T (Z - \mathbb{1}a^T) r \\ &= r^T (Z^T Z - Z^T \mathbb{1}a^T - a\mathbb{1}^T Z + a\mathbb{1}^T \mathbb{1}a^T) r \\ &= r^T (Z^T Z - 2a\mathbb{1}^T Z + Naa^T) r. \end{aligned} \tag{6}$$

This expression emphasizes the quadratic form of the objective energy w.r.t. to the $r$ variable (i.e., the versor orthogonal to the affine hyperplane). Moreover, we observe that in this expression the mean $\mu$ (or centroid) of the points $Z$ appears. Indeed, we have that

$$\mu^T = \frac{1}{N} \mathbb{1}^T Z, \tag{7}$$

thus, we get

$$E_Z(r,a) = r^T (Z^T Z - 2Na\mu^T + Naa^T) r. \tag{8}$$

By simply plugging $\mu$ in place of $a$ in (8), we get that the distance of the points $Z$ w.r.t. the affine hyperplane $L$ orthogonal to $r$ and passing through $\mu$, is

$$\begin{aligned} E_Z(r,\mu) &= r^T(Z^TZ - 2N\mu\mu^T + N\mu\mu^T)r \\ &= r^T(Z^TZ - N\mu\mu^T)r. \end{aligned} \tag{9}$$

Now, notice that the difference between $E_Z(r,a)$ and $E_Z(r,\mu)$ is always non-negative. In fact

$$\begin{aligned} E_Z(r,a) - E_Z(r,\mu) &= r^T(-2Na\mu^T + Naa^T + N\mu\mu^T)r \\ &= Nr^T(a-\mu)^T(a-\mu)r \\ &= N[(a-\mu)^Tr]^2. \end{aligned} \tag{10}$$

This fact allows us to give a lover bound of the energy $E_Z(r,a)$ in the $a$ variable, in fact we have that

$$E_Z(r,a) = E_Z(r,\mu) + N[(a-\mu)^Tr]^2 \geq E_Z(r,\mu) \tag{11}$$

for any $r \in S^d$ and $a \in \mathbb{R}^d$. As a consequence,

$$\min_{(r,a)\in S^d\times\mathbb{R}^d} E_Z(r,a) = \min_{r\in S^d} E_Z(r,\mu) \tag{12}$$

as an optimal value for $a$ in the minimization problem (4) is $a^* = \mu$. Geometrically, this means that the TLS hyperplane approximating the points $Z$ always passes through the mean of the points. We can give the following.

**Problem** (TLS equivalent formulation)**.** *Given a set of points $Z = \{z_j\}_{j=1}^N \subset \mathbb{R}^d$, find $r^* \in S^d$ such that*

$$r^* = \operatorname*{argmin}_{r\in S^d} E_Z(r,\mu). \tag{13}$$

*where $\mu$ is the mean of the points $Z$.*

A further inspection of (9) shows that $E_Z(r,\mu)$ is the quadratic function of $r$ defined by the covariance matrix of the points $Z$. Let us define $M := \mathbb{1}\mu^T$, then the usual covariance matrix of the points $Z$ is given by $C := \operatorname{cov}(Z) = (Z-M)^T(Z-M)$. Notice that

$$\begin{aligned} C &= (Z-M)^T(Z-M) \\ &= Z^TZ - 2M^TZ + M^TM \\ &= Z^TZ - 2\mu\mathbb{1}^TZ + \mu\mathbb{1}^T\mathbb{1}\mu^T \\ &= Z^TZ - 2N\mu\mu^T + N\mu\mu^T \\ &= Z^TZ - N\mu\mu^T. \end{aligned} \tag{14}$$

thus, the objective energy to be minimized $E_Z(r,\mu) = r^TCr$, is quadratic and $C = \operatorname{cov}(Z)$ is positive semi-definite.

## Eigenvector solution of the TLS problem

As we have seen, the TLS problem is a quadratic minimization problem. Its solution can be obtained in terms of eigenvectors of the covariance matrix $C$. We state here a general result.

**Lemma.** *Let $A \in \mathbb{R}^{d\times d}$ a square, symmetric and positive semi-definite matrix. The associated quadratic minimization problem $b^* = \operatorname{argmin}_{b\in S^d} b^T Ab$ is solved for $b^* = v$, where $v$ is the eigenvector of $A$ associated to the lowest eigenvalue.*

*Proof.* Matrix $A$ can be decomposed as $A = U\Sigma U^T$ where:

- $U = [u_1, \ldots, u_d]$ is orthogonal matrix, i.e., $u_j^T u_k = 0$ for $j \neq k$ and $u_j^T u_j = 1$ for all $j, k = 1, \ldots, d$.
- $\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_d^2)$ where $\sigma_1^2 \geq \ldots \geq \sigma_d^2 \geq 0$.

Of course, one has that $Au_j = \sigma_j^2 u_j$ for $j = 1, \ldots, d$. Let us consider $b \in S^d$, to prove the lemma we just need to show that $b^T Ab \geq u_d A u_d$. The set of vectors $\{u_1, \ldots, u_d\}$ forms an orthonormal basis for $\mathbb{R}^d$, so we can decompose $b$ onto this basis as $b = \sum_{j=1}^d \beta_j u_j$. By orthogonality of the basis we get

$$1 = b^T b = \sum_{j,k=1}^{d} \beta_j \beta_k u_j^T u_k = \sum_{j=1}^{d} \beta_j^2. \tag{15}$$

Therefore, we can easily see that

$$b^T Ab = \sum_{j,k=1}^{d} \beta_j \beta_k \sigma_k^2 u_j^T u_k = \sum_{j=1}^{d} \sigma_j^2 \beta_j^2 \geq \sigma_d^2 \sum_{j=1}^{d} \beta_j^2 = \sigma_d^2 = u_d^T A u_d \tag{16}$$

and the lemma is proved. □

This result provides us with a method to solve the TLS problem. The covariance matrix $C = \operatorname{cov}(Z)$ is square, symmetric and positive semi-definite. The minimization problem (13) reads exactly $r^* = \operatorname{argmin}_{r\in S^d} r^T Cr$, thus given the spectral decomposition $C = U\Sigma U^T$, we have that the solution is $r^* = u_d$. Therefore, the affine hyperplane that minimizes the sum of the square orthogonal distances of points $Z$ is orthogonal to the eigenvector of $C$ associated to the lowest eigenvalue. Moreover, it is straightforward to compute the sum of the squared distances of the points w.r.t. to the solution, in fact:

$$\sum_{j=1}^{N} \operatorname{dist}(z_j, L_{u_d,\mu})^2 = E_Z(u_d, \mu) = u_d^T C u_d = \sigma_d^2. \tag{17}$$

Therefore, if points $Z$ are randomly drawn from a given distribution, the TLS hyperplane approximating the points $Z$ is the linear subspace of $\mathbb{R}^d$ that minimizes the variance of the squared orthogonal distances of the points, i.e., the quantity $\sigma_d^2/N$ (or $\sigma_d^2/(N-1)$ for an un-biased estimator).

## Numerical example

We generate a set of random samples from a 3-variate Gaussian distribution. The covariance matrix is diagonal with decreasing values on the diagonal. Thus, the 3-rd axis will be the direction orthogonal to the TLS plane approximating the points and the smallest eigenvalue is an approximation of the variance of the points in the direction of the 3-rd axis.

```
%generate N 3-variate Gaussian random samples with mean MU and
%diagonal covariance matrix SIGMA
N = 500;
MU = [5,5,5]';
SIGMA = [4^2,0,0;0,2^2,0;0,0,1^2];
Z = mvnrnd(MU,SIGMA,N);

%solve the TLS problem by finidng the eigenvector V associated
%to the smallest eigenvalue of the covariance matrix of the
%points
C = N*cov(Z,1);
[V,D] = eigs(C,1,'sm');
%the vector orthogonal to the TLS plane is approximately
%parallel to the 3-rd axis
V =
    0.0062
    0.0269
   -0.9996

%the variance of the squared orthogonal distances
%of the points w.r.t. to the TLS plane approximates the
%lowest variance in the SIGMA matrix
mu = mean(Z)
    4.9123    4.9133    5.0634
var = D/N
    1.0097
```